IN4MATX 133: User Interface Software

Lecture 17:
Generative AI and
Interface Development

Today’s goals

By the end of today, you should be able to…

- Explain, at a high level, how Large Language Models respond to prompts
- Follow a few best practices for prompting, particularly when coding
- Articulate opportunities for LLMs to support interface development and design practices
- Recognize the limits of LLMs towards supporting those practices

Disclaimers

- This is an overview of how LLMs and GenAI work, and how that influences when you might and might not want to use them for creating interfaces
  - I can’t cover the entirety of LLMs and GenAI in one lecture
- The field is still emerging, much of what I say today may be irrelevant in 5 years

What’s a Large Language Model, and how does it work?

Taking a step back: Natural Language Processing

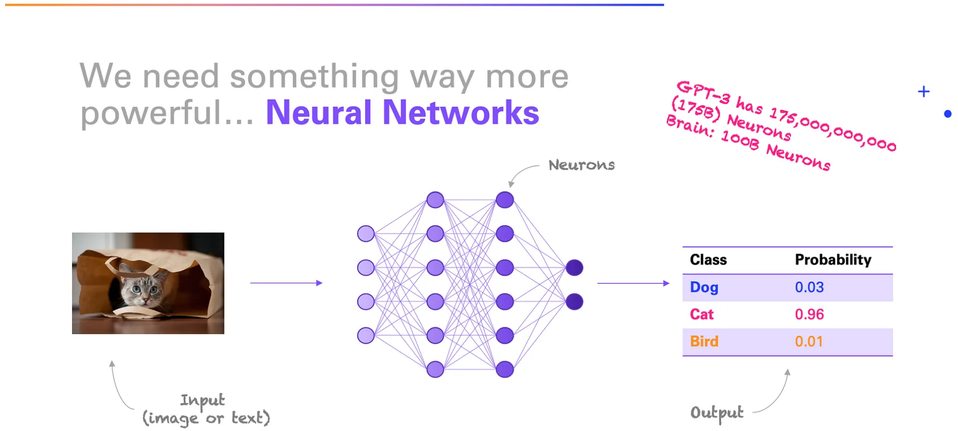
Natural Language Processing (NLP)

- Super common NLP task: predict the next word in a sentence
  - “The cat likes to sleep in the _____”
- How do we make this prediction?
  - We have a dictionary. We know what words are theoretically possible, but some will be more likely than others
  - So we get a bunch of example sentences, and try to model what might come next
  - Given enough examples, we can make good predictions

https://medium.com/data-science-at-microsoft/how-large-language-models-work-91c362f5b78f

Natural Language Processing (NLP)

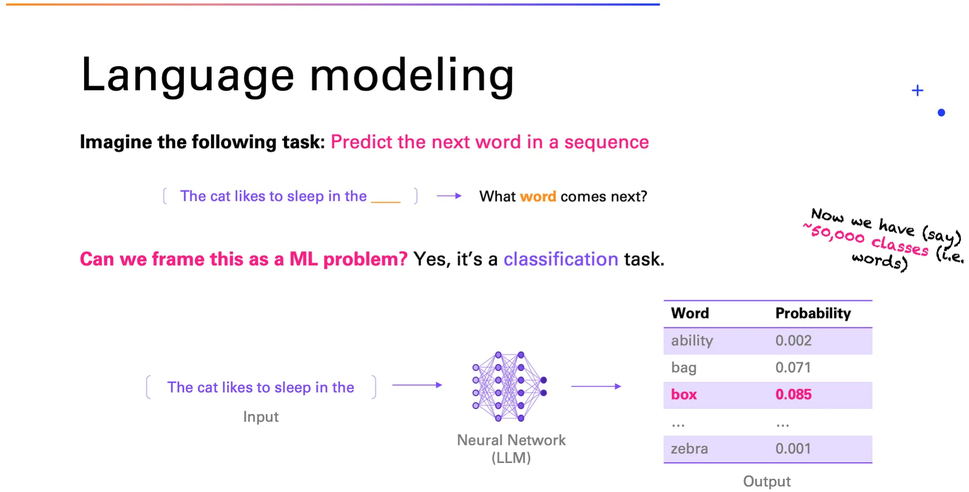
- This is where math comes in, which I'll mostly gloss over
- Computers try to act like brains, with tons of "neurons" that activate when they see a pattern they've seen before
- E.g., we have lots of example sentences which include both "cat" and "box"



https://medium.com/data-science-at-microsoft/how-large-language-models-work-91c362f5b78f

Natural Language Processing (NLP)

- This is still prediction, with a probability associated
  - E.g., 8.5% of the time we will say that the cat likes to sleep in the “box”, 7.1% in the “bag”



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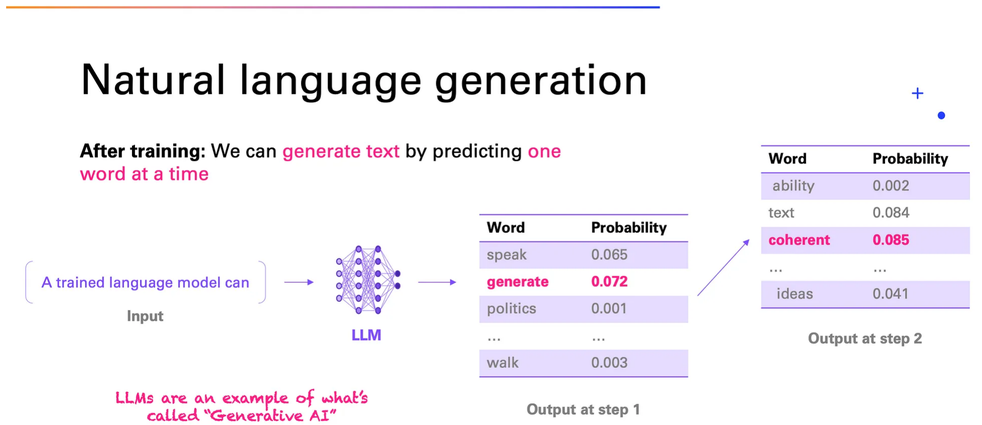
Natural Language Processing (NLP)

- If we chain this together, one word at a time, we can start to generate sentences



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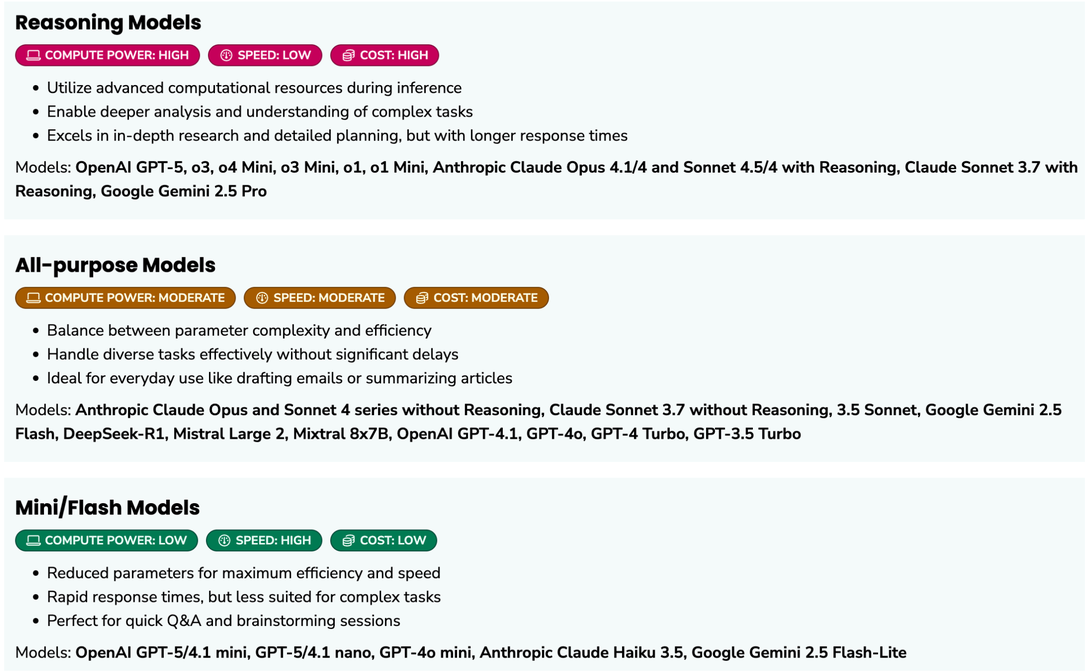
Large Language Models

- What I've described is NLP. What makes *Large Language Models* special?
- They're **Large**, in two ways
  - They have a ton of input examples. Like, the entire internet, and everything that's ever been digitized (books, media, etc.). So they know a lot about a lot of things
  - They have a ton of neurons. Like, more than the human brain, which lets them do really good pattern matching
- This poses good questions of copyright, ethics, etc.

Large Language Models

- Models can vary in how “large” they are (neurons, input)
- Larger models:
  - Offer more advanced the reasoning, theoretically
  - Require more compute power
  - Cost more
  - Respond more slowly



https://zotgpt.uci.edu/learn/models/

Question 

From an environmental perspective, which model is more sustainable?

A. Larger models

B. Smaller models

C. They will be about the same

D. I’m not sure

E. [space intentionally left blank]

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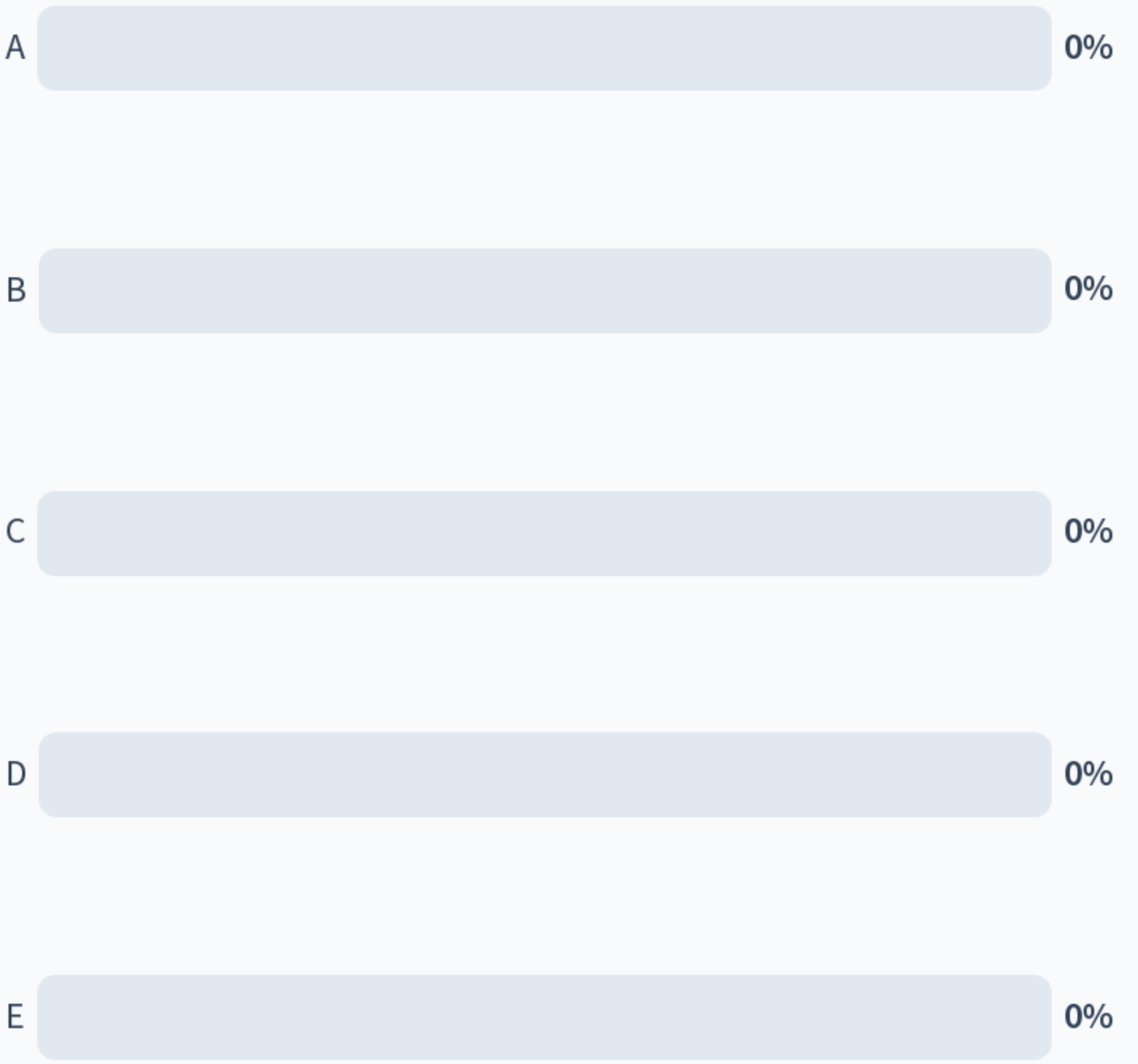

Question 

From an environmental perspective, which model is more sustainable?

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D. I’m not sure

E. [space intentionally left blank]

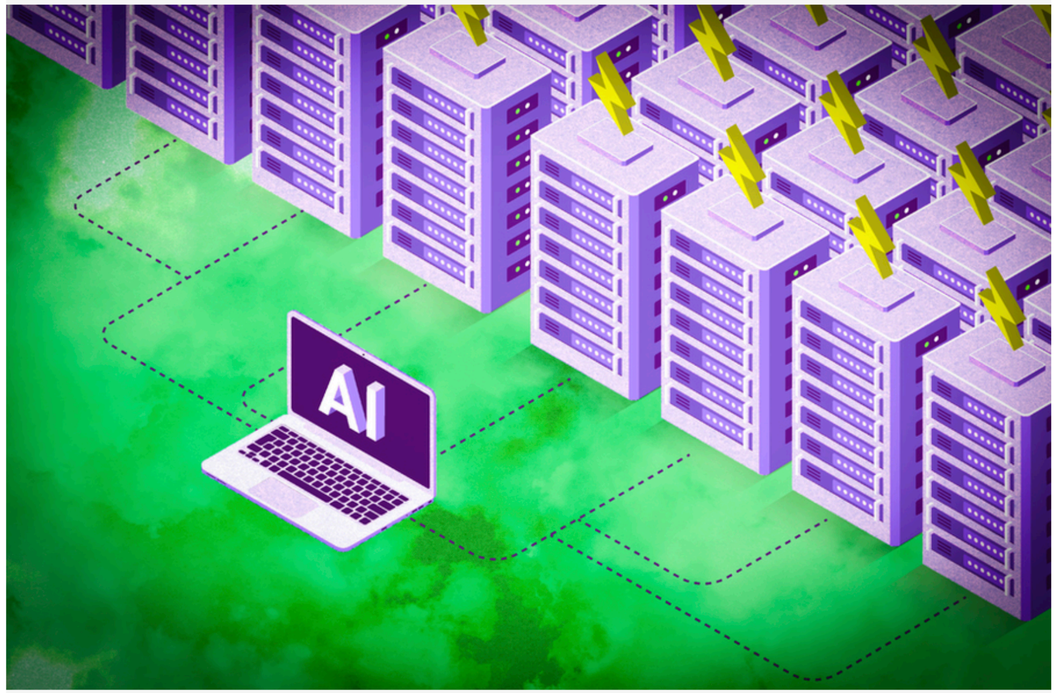
Large Language Models

- The impact of Generative AI on the environment is massive
  - Data centers to train models and respond to queries demand electricity and cooling
  - Models are constantly updated, so energy to train prior versions gets wasted
  - Computation demands top-of-the-line hardware and needs refreshing, resulting in hardware waste

Explained: Generative AI's environmental impact

Rapid development and deployment of powerful generative AI models comes with environmental consequences, including increased electricity demand and water consumption.

Adam Zewe | MIT News
January 17, 2025



https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117

Large Language Models versus Generative AI

LLMs vs. GenAI

- Large Language Models are specifically trained to understand and generate text
  - Examples: ChatGPT, Gemini, Claude
- Generative AI is a broader category of using AI to generate other forms of content
  - Images/Videos: Mindjourney, Stable Diffusion
  - Code: Copilot, Replit, Cursor

LLMs vs. GenAI

- Generative AI may have non-text input and/or non-text output, but the fundamentals are the same
  - E.g., if we want to generate images, we still process example images in largely the same way
  - Same for code

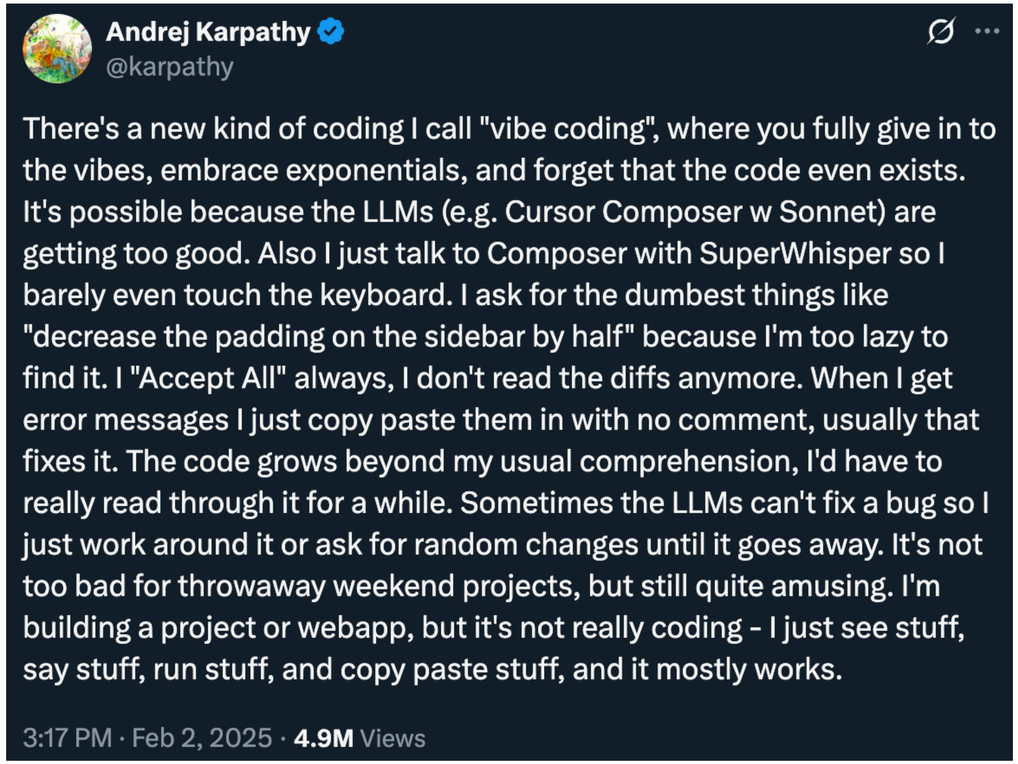
So how do LLMs/GenAI help with Interface Development? Design?

GenAI and Interface Development

- Tons of examples of code exist, so Generative AI can similarly be used to generate code which meets certain criteria
- What can get generated well?
  - Common functionality: If it's likely that someone has done it before, an LLM can help you reproduce it
  - Syntax: LLMs pick up patterns in the examples, code is strict in syntax, therefore lots of examples

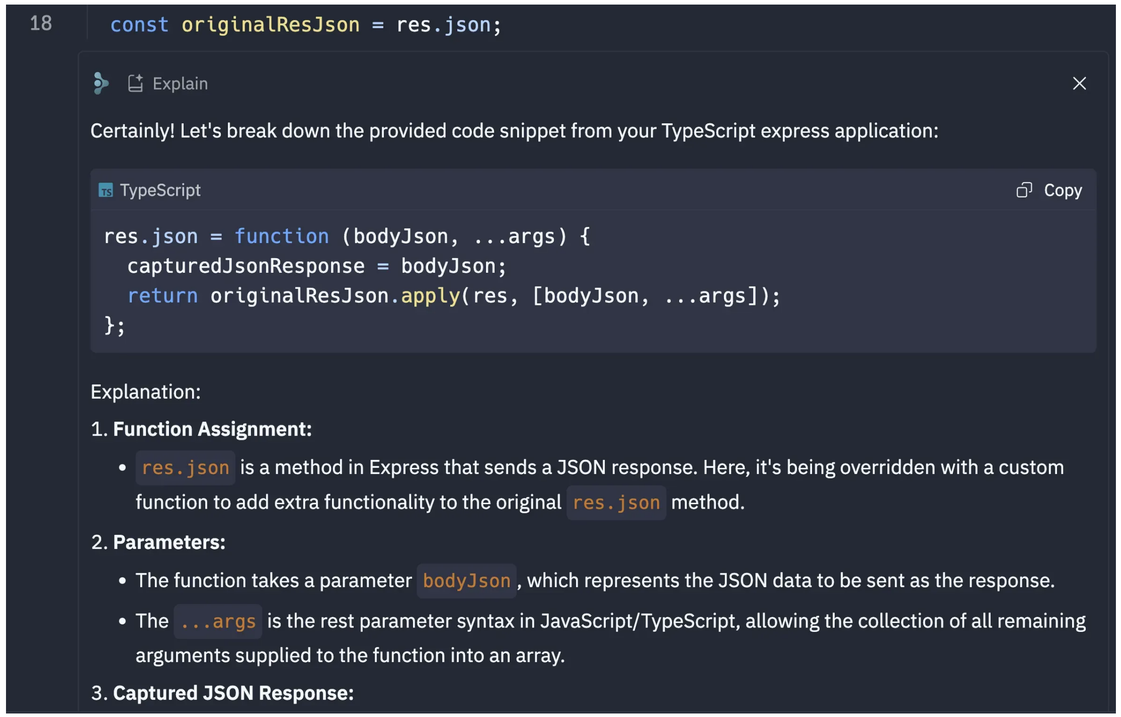
GenAI and Development

- “Vibe coding”: the latest craze
  - Ask an LLM to make a thing, ask it to fix the errors
- “Not too bad for throwaway weekend projects”



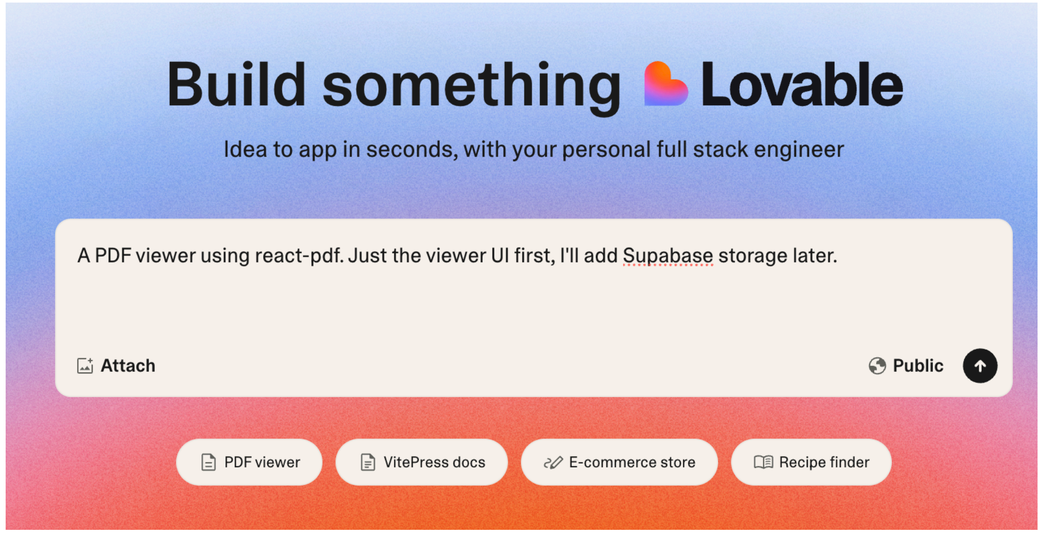
GenAI and Development

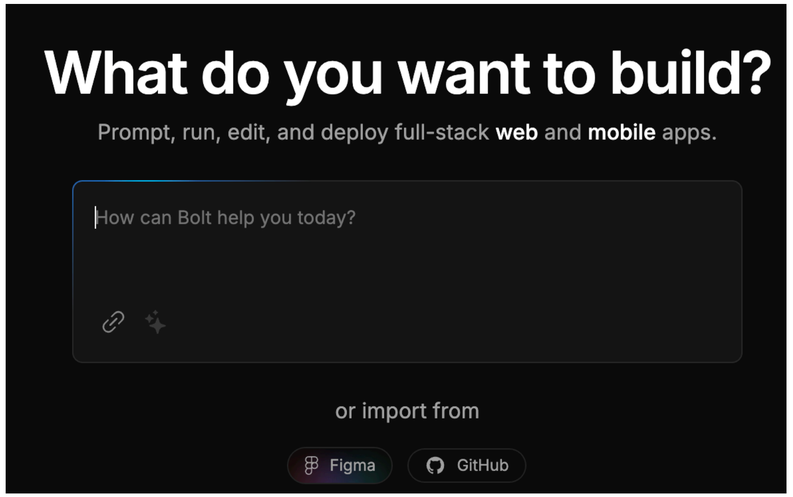
- Copilot, Replit, Cursor all can help with generating code that fulfill specific goals
  - Drawing on examples that exist
- And, can attempt to explain what a piece of code does
  - Drawing on documentation that exists
- “Better autocomplete”, “Improved documentation”



GenAI and Design

- Tools exist, or will exist, to help implement apps via text descriptions and/or convert Figma mockups to functional prototypes
- Loveable, Bolt, Figma Make
- Worth looking into and seeing what you think of them

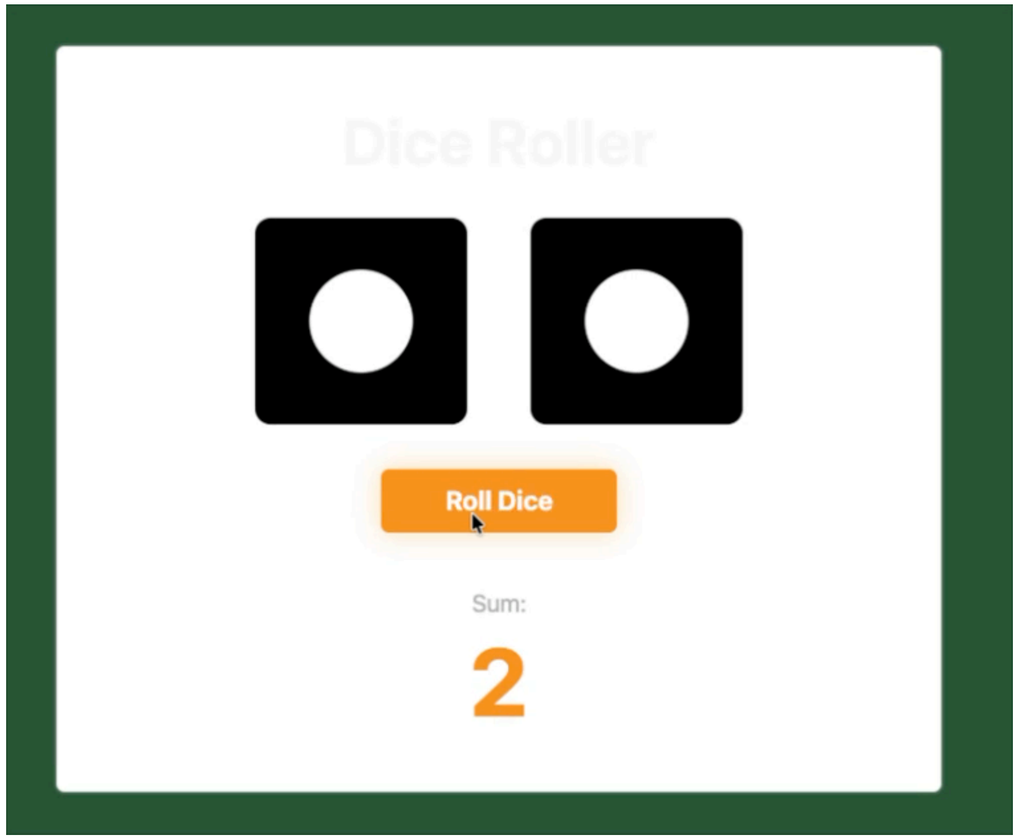




https://lovable.dev/
https://www.figma.com/make/
https://bolt.new/

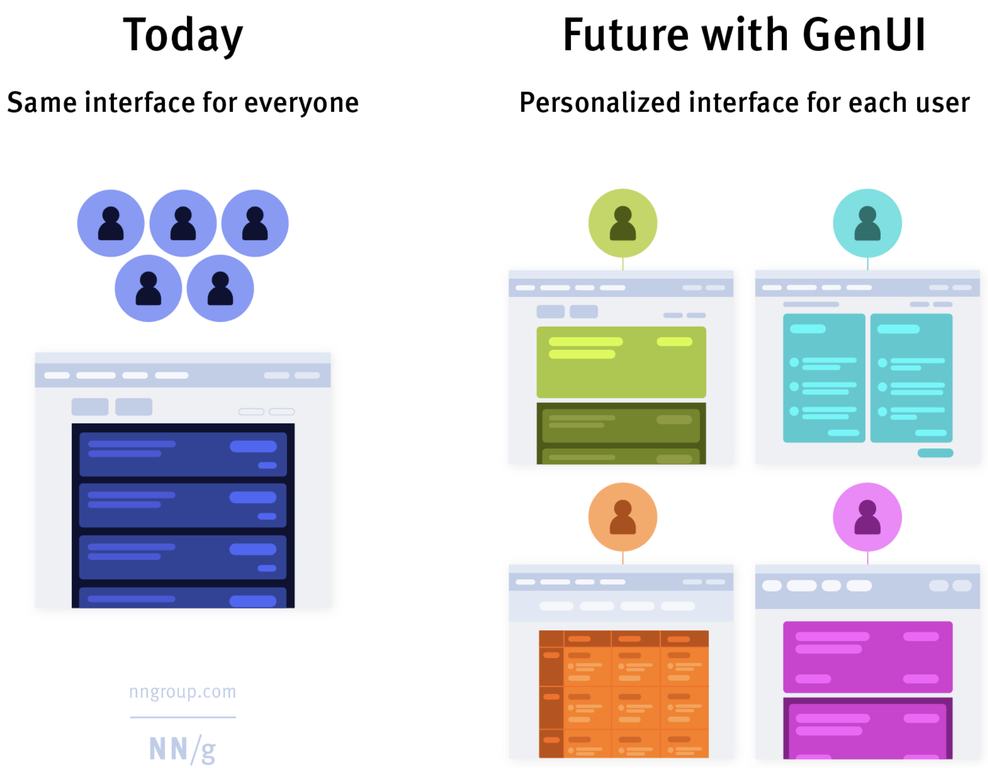
Let’s try it!

https://lovable.dev/



Generative UI

- People are also speculating that eventually, AI will enable creation of personalized interfaces that are customized to fit user's needs and contexts
  - More accessible designs, maybe?
- For now, just a vision



https://www.nngroup.com/articles/generative-ui/

Effective prompting

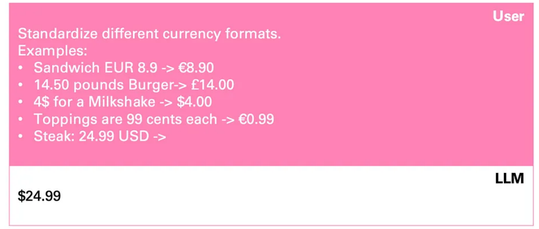
Effective prompting

- Providing examples
  - “Write me some code that turns X input into Y output”
  - “Format phone number 9492049299 as (949) 204-9229”
- “Few shot” learning

Few-Shot Learning

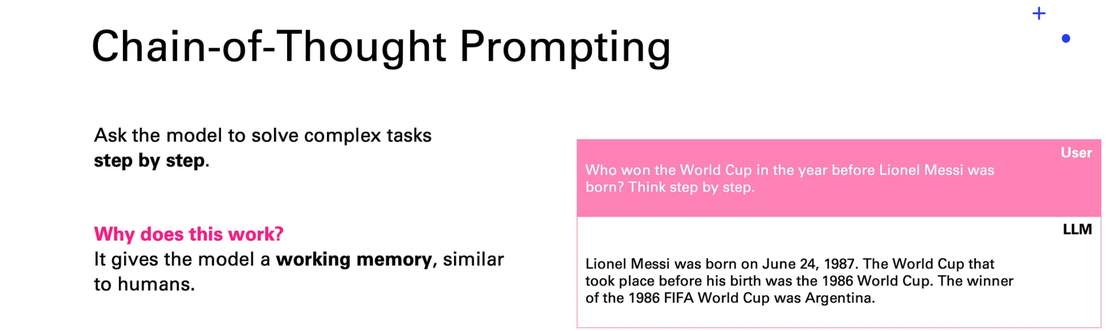
Providing **examples** helps the LLM understand and follow your task.

This is especially helpful to ensure a specific **output format**.



Effective prompting

- Asking for step-by-step thinking
  - “Make an app an interface with XYZ characteristics. Then, connect it to a database that has ABC columns”
  - Could break it down into separate prompts
- Chain-of-thought prompting



Effective prompting

- Overall, recognize that prompting is a conversation
  - Ask for edits, refinement, etc.
- Offer context when you can
  - Everything from low-level stuff like programming language to higher-level design goals
  - Specificity helps, but LLMs can also get overwhelmed
- Ask for explanations to help you understand what was produced

Effective prompting

- I think/hope that we will introduce other interaction paradigms eventually, but for now prompting is what we have
  - Research is actively working on this
  - We have a lot of research suggesting that text is not great for these sorts of inputs

Limitations

Missing context

- GenAI only knows what you tell it, and problems are often under-specified
  - GenAI doesn’t know the rest of your codebase
  - GenAI doesn’t know your design context
  - As you feed in more context via your prompts, the predictive power decreases
- What’s really tough about coding is piecing together smaller functions, algorithms, etc. to work as you hope them to
  - Also, connection between different parts of your application like the interface, database, authentication flow, etc.

Question

Among: AP Calculus BC, AP US History, and AP English Literature, which does ChatGPT (4.0) do best at? Worst at?

A. Best at Calculus, worst at History

B. Best at Calculus, worst at English

C. Best at History, worst at Calculus

D. Best at History, worst at English

E. Best at English, worst at Calculus

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	GPT 4.0	GPT 3.5
AP US History	5 89th–100th	4 74th–89th
AP Calculus BC	4 43rd–59th	1 0th–7th
AP English Language and Composition	2 14th–44th	2 14th–44th
AP English Literature and Composition	2 8th–22nd	2 8th–22nd

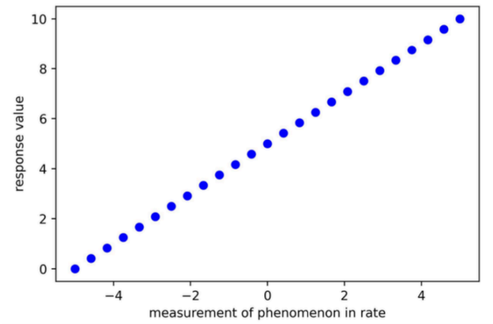
https://openai.com/index/gpt-4-research/

Hallucinations/truthiness

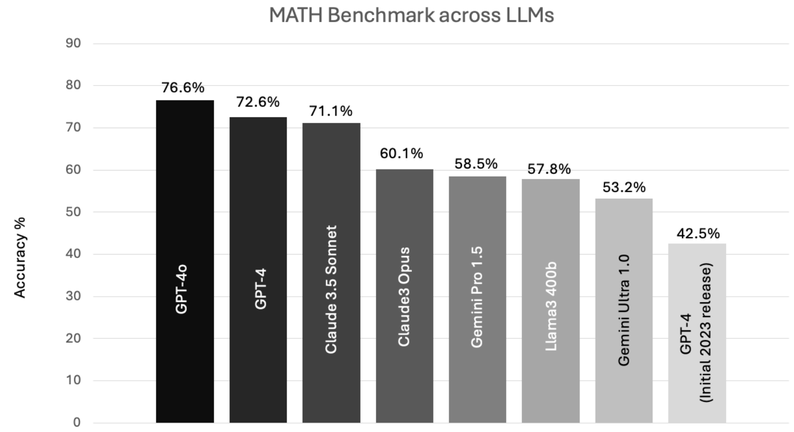
- LLMs are, inherently, probabilistic. They're predicting the next word, then choosing among the options
- There is nothing to ensure that LLMs are producing text which is factual
  - And, if some text on the internet says something with confidence, LLMs will repeat it
- These are *hallucinations*, LLMs making up facts which aren't true
  - Potentially solvable by adding trusted sources and comparing against them
  - But still an active research area

Hallucinations/truthiness

- For example: LLMs are surprisingly bad at math, even basic math
  - LLMs are building off of examples, rather than a deeper understanding of the context
  - ChatGPT etc. will probably get 1+1=2 correct, because it's seen examples, but screw up other basic math problems
- This is also an active area of research, and has improved a lot

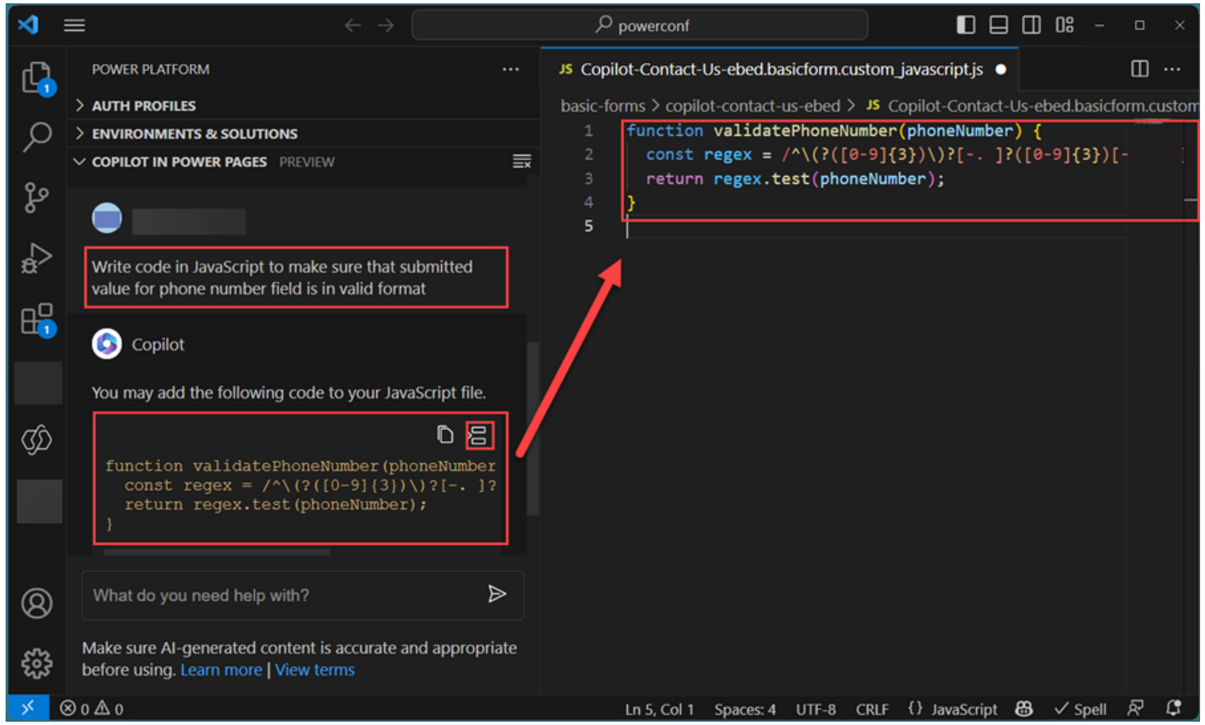


What is the minimum x value among the set of points in the figure?	4 ✗
What is the minimum value for the range on the x axis?	0 ✗



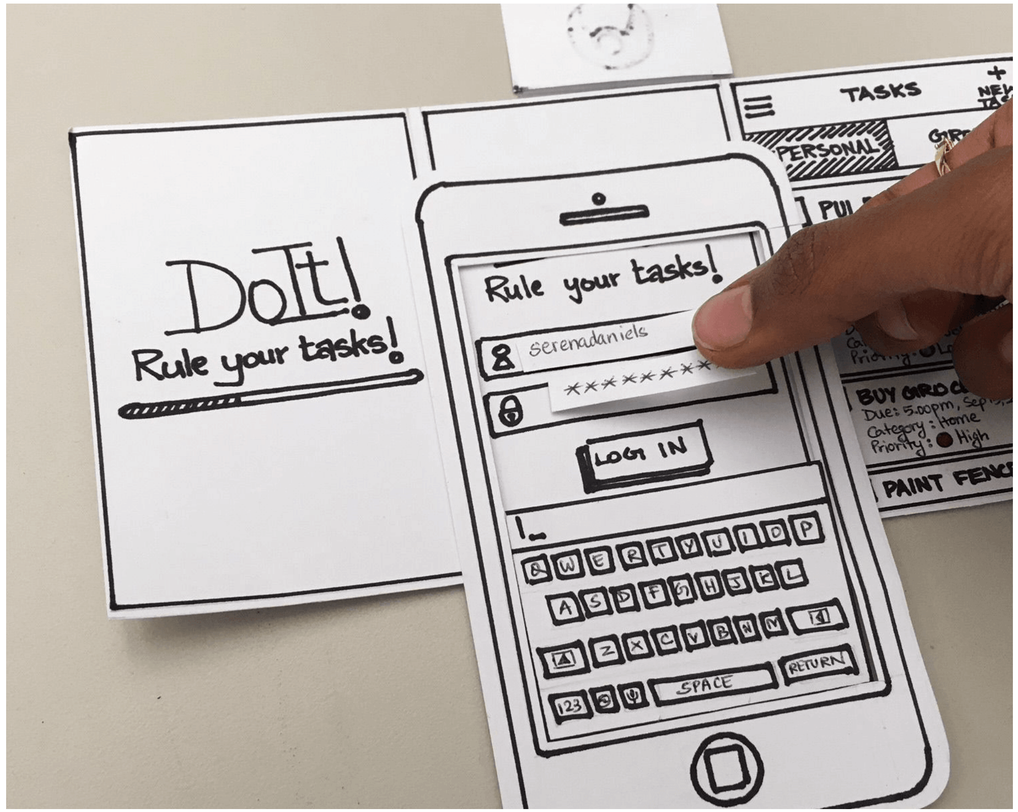
Incomprehensible

- Code generation is one thing, but developers need to understand what the code is doing
  - Is it introducing bugs? Errors? Security vulnerabilities?
  - Is it inefficient? Requiring libraries which aren't needed?
- Does using GenAI make developers less diligent? Maybe



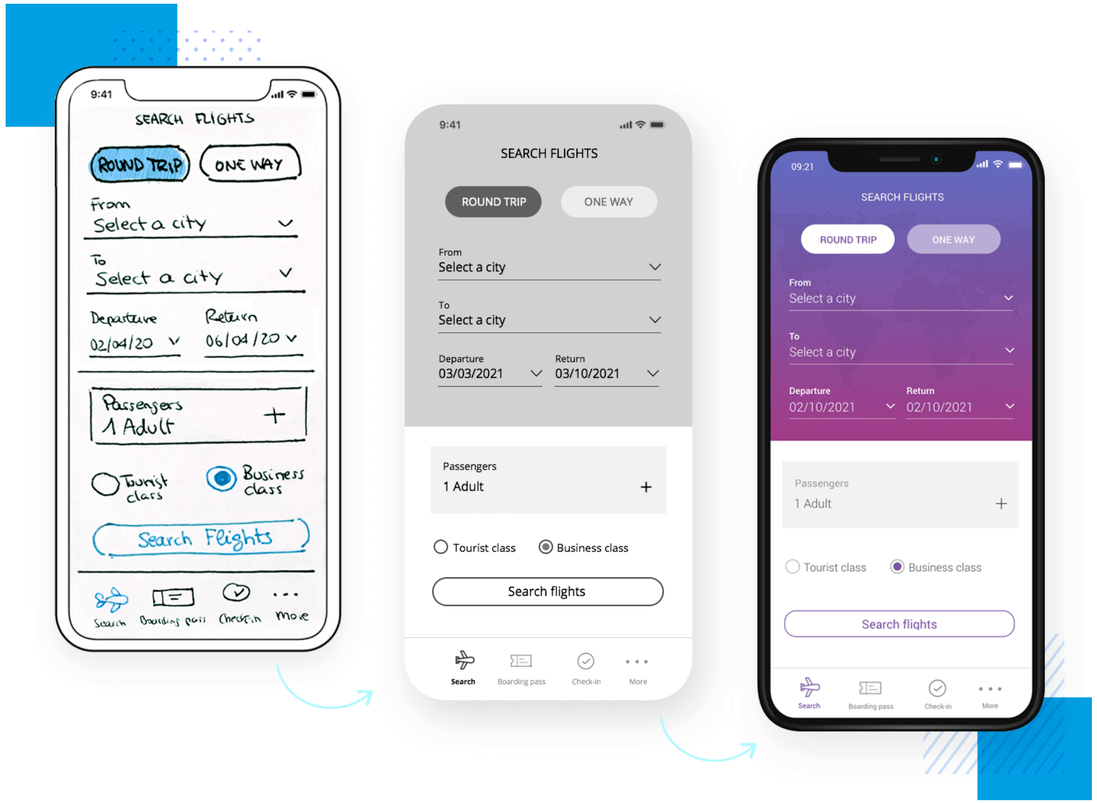
Perceived polish

- If you've taken IN4MATX 131/132, you're familiar with low-fidelity and high-fidelity prototypes
- Low-fidelity prototype: a sketch to test out a potential interaction or design idea
  - How should login functionality work?
  - How should our pages be organized or ordered?
  - Useful for getting feedback



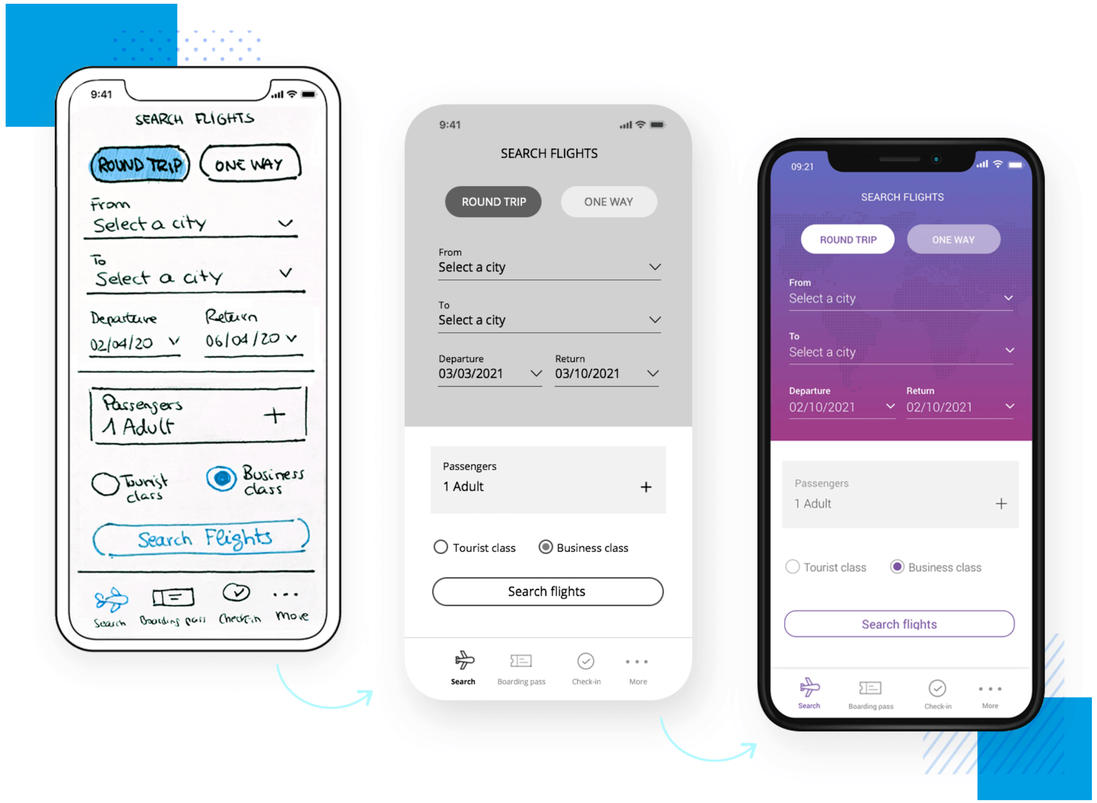
Perceived polish

- Traditionally, low-fidelity prototypes are faster to make
  - You sketch them on paper or similar to reduce the burden
  - Once you feel good about the interaction, you move onto higher-fidelity prototyping
  - A byproduct, your design looks more polished as you improve it



Perceived polish

- … But with GenAI, you can easily make interfaces which *look* polished (high-fidelity) even when the interaction isn't
  - And if they look polished, people's feedback will be on the polish
- Therefore, you have to be more intentional about doing your prototyping



Privacy and IP

- Anything that you connect to GenAI becomes part of the examples it uses for training
  - Your intellectual property can become part of the example set
  - Private user data, etc.
- There are ways of avoiding this risk, like enterprise agreements. But it’s worth discussing and keeping in mind

Privacy and IP

- But, if you pay, models can be federated to prevent data sharing
- ZotGPT is a good example of this

What level of privacy is available with ZotGPT? Who can see my chats?

- **For ZotGPT Chat:** Your chats are not used by Microsoft, UCI, or OpenAI to train AI models. Your chats are private and secure. You can delete your chat history at any time.
- **For Microsoft Copilot:** Your chats are not used by Microsoft, UCI, or OpenAI to train AI models **as long as you authenticate using your UCI credentials**. Your chats are confidential.
- **For Google Gemini:** Your chats are not used by Google to train the Gemini model **as long as you authenticate using your UCI credentials**.

Learn more on the ZotGPT Privacy Notice.

What type of data is safe to use with ZotGPT?

	P1	P2	P3	P4 / HIPAA
ZotGPT	✅ Yes	✅ Yes	✅ Yes	❌ No
UCI Microsoft Copilot	✅ Yes	✅ Yes	✅ Yes	❌ No
UCI Google Gemini	✅ Yes	✅ Yes	❌ No	❌ No

Learn more about Data Protection Levels

https://www.oit.uci.edu/services/ai/zotgpt

Overall reflections

Overall reflections

- Generative AI can be decently effective at entry-level coding and design
  - I wouldn’t be surprised if it did a good job at our class assignments, for example
- But, you have to know what to ask it to do
  - Having domain expertise and design/technical expertise helps

Overall reflections

- But, use with caution
  - Using it changes your work from generating to stitching/fixing
  - Is this less work? Easier work? Depends.
- I don’t see UX designers or developers getting replaced by AI any time soon
- **What have your experiences been?**

Today’s goals

By the end of today, you should be able to…

- Explain, at a high level, how Large Language Models respond to prompts
- Follow a few best practices for prompting, particularly when coding
- Articulate opportunities for LLMs to support interface development and design practices
- Recognize the limits of LLMs towards supporting those practices

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